Class XII Session 2025-26
Subject - Physics
Sample Question Paper - 9

Time Allowed: 30 Minutes

General Instructions:

1. There are 33 questions in all. All questions are compulsory.
2. This question paper has five sections: Section A,
3. All the sections are compulsory.
4. Use of calculators is not allowed.

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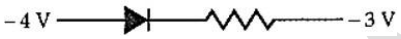
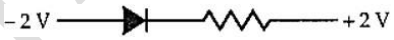
Section A

1. A forward biased diode is **[1]**

a) 

b)

c)

d) 

2. A cell of constant emf is first connected to a resistance R_1 and then connected to a resistance R_2. If power delivered in both cases is same, then the internal resistance of the cell is: **[1]**

a) $\sqrt{R_1 R_2}$

b) $\sqrt{\frac{R_1}{R_2}}$

c) $\frac{R_1 - R_2}{2}$

d) $\frac{R_1 + R_2}{2}$

3. Magnifying power of a compound microscope is high if **[1]**

a) both objective and eye-piece have long focal lengths

b) the objective has a long focal length and eye-piece has a short focal

c) the objective has a short focal length and the eye-piece has a long focal

d) both objective and eye-piece have short focal lengths

4. Which of the following has its permeability less than that of free space? **[1]**

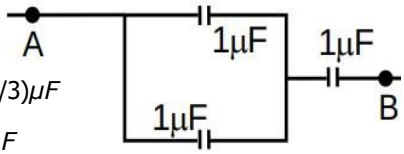
a) Copper chloride

b) Nickel

c) Aluminium

d) Copper

5. Three capacitors each of $1\mu F$ are connected as shown in figure. The capacitance between the point A and B: **[1]**



a) $(2/3)\mu F$
b) $(1/3)\mu F$
c) $1\mu F$
d) $3\mu F$

6. If electron velocity is $2i + 3j$ and it is subjected to magnetic field of $4k$, then its **[1]**

a) path will change
b) speed will change
c) path will not change
d) both path will change and speed will change

7. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon: **[1]**

a) the rates at which currents are changing in the two coils
b) the material of the wires of the coils
c) the currents in the two coils
d) relative position and orientation of the two coils

8. The force between two magnetic poles is F. If the distance between the poles and pole strengths of each pole are doubled, then the force experienced is: **[1]**

a) $\frac{F}{4}$
b) 2F
c) F
d) $\frac{F}{2}$

9. The refractive index of glass is 1.5. Which of the two colors red or violet travels slower in a glass prism? **[1]**

a) both red and violet travel slower at the same speed
b) red component of white light travels slower than the violet component
c) both red and violet travel faster at the different speed
d) violet component of white light travels slower than the red component

10. A point charge of $100\mu C$ is placed at $3i + 4jm$. Find the electric field intensity due to this charge at a point located at $9i + 12jm$: **[1]**

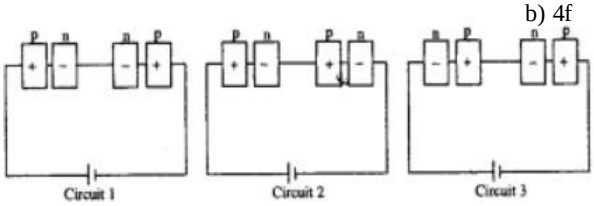
a) $4500Vm^{-1}$
b) $8000Vm^{-1}$
c) $9000Vm^{-1}$
d) $2250Vm^{-1}$

11. Two identical p-n junctions may be connected in series with a battery in three ways as shown. The potential drops across the two p-n junctions are equal in **[1]**

a) circuit 3 only
b) circuit 1 only
c) circuit 1 and 2
d) circuit 2 and 3

12. The minimum distance between an object and its real image formed by a convex lens of focal length f is: **[1]**

a) 3f
b) 4f



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c) 2f

d) f

13. **Assertion (A):** The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energies. **[1]**

Reason (R): The energy of electrons emitted from inside the metal surface, is lost in collision with the other atoms in the metal.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion (A):** Electron move away from a region of lower potential to a region of higher potential. **[1]**

Reason (R): An electron has a negative charge.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

15. **Assertion (A):** In Young's double slit experiment the fringes become indistinct if one of the slits is covered with cellophane paper. **[1]**

Reason (R): The cellophane paper decreases the wavelength of light.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. A resistor, a capacitor, and an inductor are connected in series. The combination is connected across an ac source of frequency 50 Hz. **[1]**

Assertion (A): Peak current through each remains the same.

Reason (R): Average power delivered by source is equal to average power developed across the resistance.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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